Dishari: AI-Powered Smart Navigation System for the Visually Impaired

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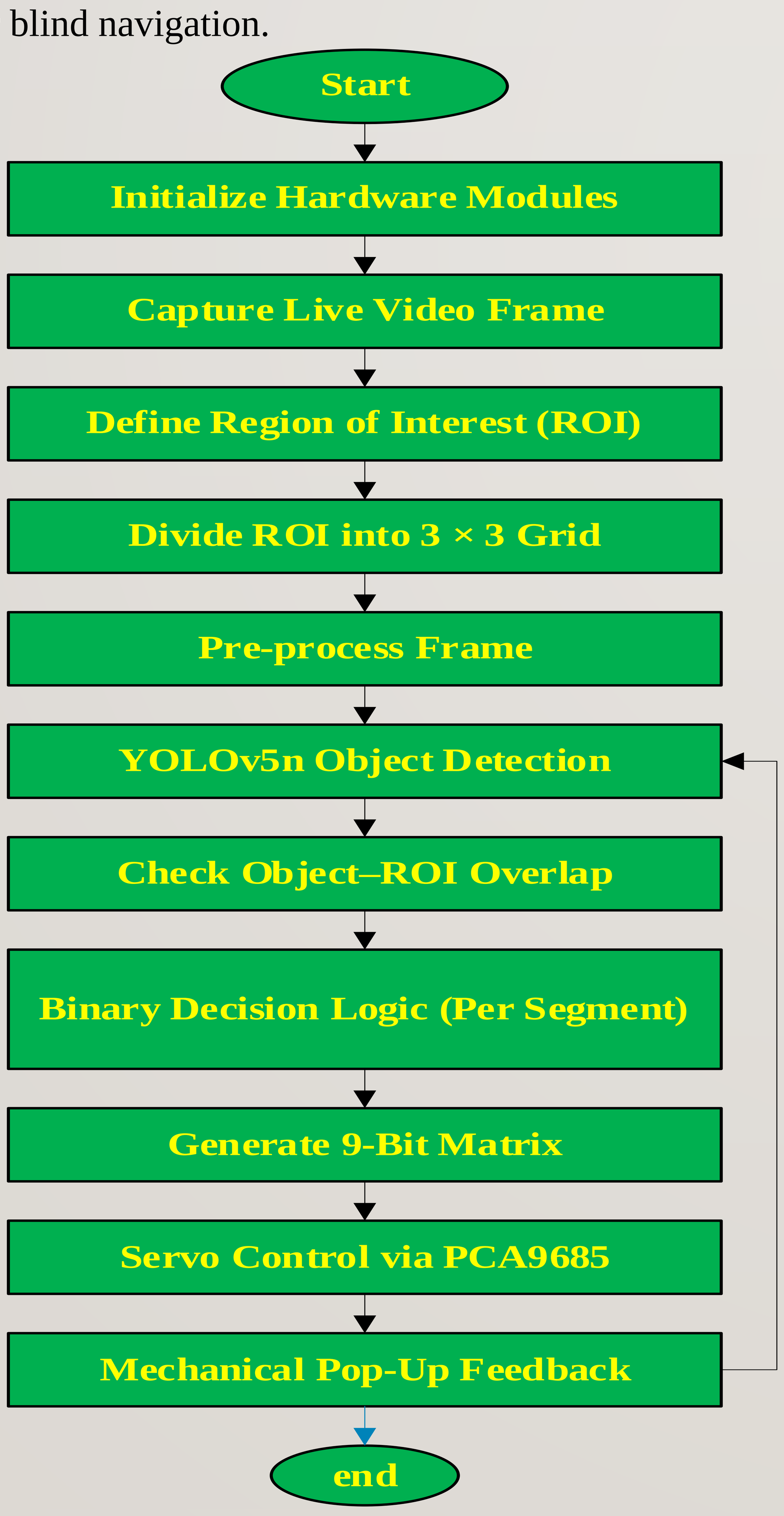
ABSTRUCT

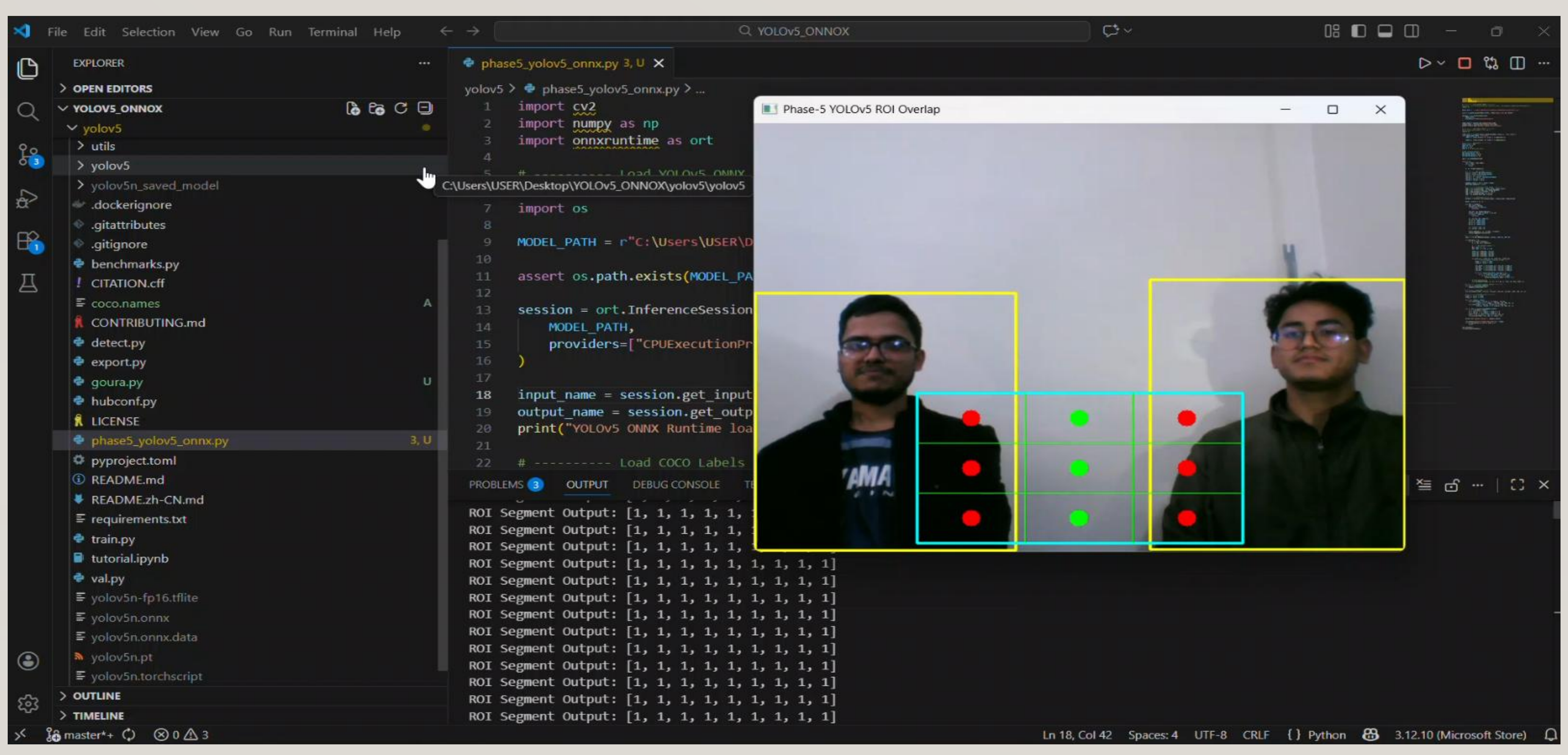
Independent navigation remains a major challenge for visually impaired individuals, particularly in outdoor environments filled with moving people, vehicles, and other obstacles. This project offers a real time vision-based assistive system for the visually impaired using Raspy PI 4B, where images of what lies ahead is captured continuously from a Pi camera and an extremely lightweight deep learning model (YOLOv5-nano) detects meaningful high-priority obstacles (such as people, vehicles animals and objects commonly found at side-walks- beds / shrubbery also nonobstructive elements can be ignored in background). To spatially localize obstacles, the bottom-middle ROI (which represents user walking path) is further divided into a 3 × 3 grid. Each grid segment generates a binary output indicating obstacle presence, which directly controls a corresponding servo motor via a PCA9685 driver. The servo motion is mechanically converted into a tactile pop-up feedback system, enabling intuitive, real-time obstacle awareness without audio dependency. The system operates fully on-device, offering low latency, portability, and suitability for real-world assistive navigation applications.

METHODOLOGY

The system uses a Raspberry Pi 4B with a Pi Camera to capture real-time video. A bottom-middle region of interest (ROI) on the frame is divided into a 3×3 grid, representing the walking path. Each frame is processed by a YOLOv5n ONNX model to detect relevant obstacles such as people, vehicles, animals, and objects, while ignoring background elements like sky or road. Detected objects overlapping any grid segment trigger a binary signal (1 for obstacle, 0 for empty), which is sent to a PCA9685-controlled servo for each segment. The servos rotate to provide a pop-up tactile feedback for the user, holding the position while obstacles remain present. This loop runs continuously, offering real-time assistive guidance for blind navigation.



EXPECTED RESULT



INTRODUCTION

Obstacle detection is a major challenge for visually impaired people in independent mobility, especially in dynamic outdoor environments. This project describes a real-time assistive vision system based on deep learning-based object detection for obstacle identification in the walking path and implemented on a Raspberry Pi 4B. A YOLOv5-Nano model detects relevant objects inside a defined region of interest and divides it into a 3 × 3 grid. Each grid segment outputs a binary signal indicating the presence of an obstacle and activates a corresponding pop-up mechanism driven by a servo motor for an intuitive tactile feedback. The system is compact, low-cost, and able to work in real-world environments, making it suitable as an assistive mobility aid for visually impaired users.

OBJECTIVES

I. To develop a low-cost portable assistive device for real-time obstacle detection to facilitate the safe mobility of the visually impaired users.
II. To provide faster and safer navigation by prioritizing obstacle detection in critical regions along the user's walking path.
III. Provide intuitive tactile and/or auditory feedback for improved spatial awareness without adding cognitive load.
IV. To have low power consumption, real time, portability through embedded system optimization.
V. To evaluate the system performance in indoor and semi-outdoor environments in terms of accuracy, response time and usability.
VI. To increase user confidence, reduce the risk of collisions and encourage independent navigation.

SYSTEM SPECIFICATIONS & DESIGN

1. System Specifications
2. Processing Unit: Raspberry Pi 4B (8GB RAM, 64GB microSD)
3. Camera Module: Raspberry Pi Camera v2 / Picamera2 (RGB, 640×480)
4. AI Model: YOLOv5n (ONNX, CPU inference)
5. Software: Python 3, OpenCV, ONNX Runtime, Picamera2
6. ROI Grid: Bottom-middle, 3×3 segments
7. Obstacle Detection: Person, Bicycle, Car, Bus, Truck, Chair, Table, Animal, Pole, Box, Bag
8. Servo Actuation: 9× SG90/standard servos via PCA9685 driver
9. Feedback Mechanism: Rotary → Linear pop-up for tactile obstacle indication
10. Frame Rate: ~8–12 FPS (real-time)
11. Power Supply: 5V 3A (for Pi + servo motors)

SCAN HERE

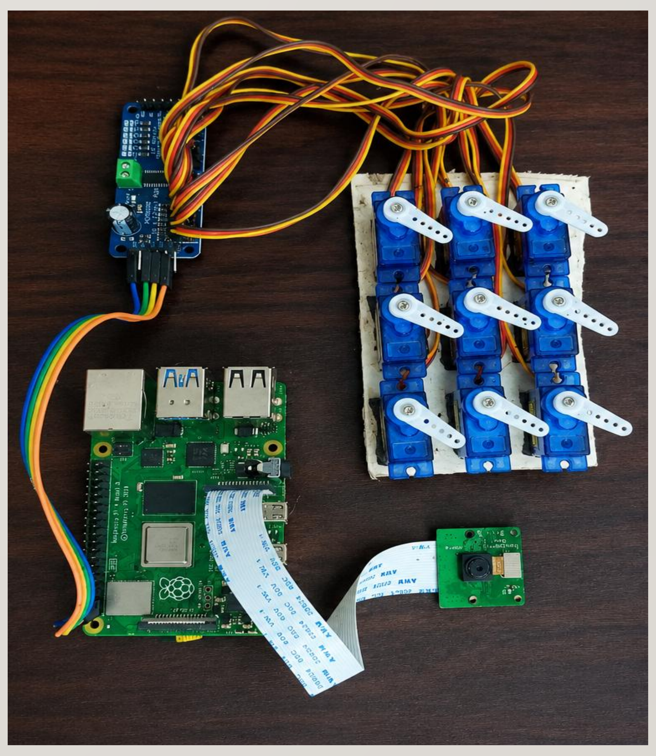


Figure 1: Experimental Setup

DISCUSSION & CONCLUSION

This project presents a real-time vision-based assistive system designed to support independent navigation for individuals with visual impairments. The system is built using a Raspberry Pi 4B, a Pi camera, and the YOLOv5n object detection model to continuously monitor the user's surroundings. It detects potential obstacles, including pedestrians, vehicles, animals, and common roadside objects, within a designated walking-path region. To determine the obstacle's location, the monitored area is divided into a 3×3 grid, and the detected information is translated into tactile feedback through servo-driven pop-up indicators. By converting visual information into intuitive physical cues, the system helps users become more aware of obstacles in their path without relying on audio feedback. The use of ONNX-based inference enables efficient and responsive performance on the Raspberry Pi, allowing the system to operate in real time while maintaining reliable detection accuracy. Overall, the results demonstrate that a low-cost embedded platform can effectively improve environmental awareness and mobility for visually impaired individuals, highlighting its potential as a practical, accessible, and scalable assistive technology solution.